Observable Quotients and Entropy Factorization

Universal Operational Reduction, Fixed-Protocol Geometry, Maximal Admissibility, and Exact Entropy Decomposition

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Abstract

A declared experiment generally fails to distinguish all states of a latent presentation. We define equality of complete declared observation laws and form the observational quotient $Q_{\mathcal{E}} = \Theta/\sim_{\mathcal{E}}$. The first main theorem proves that $Q_{\mathcal{E}}$ is the universal and maximal operational state object: every experiment-internal construction, and only such a construction, factors uniquely through the quotient map. Hidden extensions, state splittings, and other latent refinements preserving the complete observable law are therefore unidentifiable from the experiment.

The quotient alone does not select a metric, a resolution threshold, nuisance conventions, stability requirements, coarse-graining rules, or transport. These enter through a declared protocol Γ. On a regular finite-dimensional classical fibre, established Fisher–Rao characterization, spectral functional calculus, greatest-subobject maximality, and cocycle gluing yield a fixed-protocol representation theorem. In the linear case the final observable sector is

$$W_{\mathrm{obs|adm}} = \operatorname{im} P_\tau \cap N^\perp \cap S \cap C \cap T,$$

and is unique by simultaneous soundness and completeness. There is no protocol-free uniqueness claim.

Entropy is then placed strictly downstream of a quotient and a probability law. For a discrete microstate X and deterministic quotient $Q = \pi(X)$, the exact identity

$$H(X) = H(Q) + H(X \mid Q)$$

separates uncertainty between observable classes from uncertainty inside observational fibres. If the conditional law in a fibre is uniform with multiplicity $\Omega(q)$, its fibre entropy is $\log \Omega(q)$; this is distinct from the Shannon entropy of the random quotient. A conditional thermodynamic bridge is proved only when the observational partition coincides with the physical macrostate partition and the measure is thermodynamically appropriate.

Finite numerical support is reported as an attack ledger rather than as proof. In an exhaustive six-mode hard-spectral battery, all 4096 projector/sector competitors reduce to 64 after threshold consistency, 2 after gate soundness, and exactly 1 after completeness; 128 random oblique projector attacks produce no survivor satisfying all projector axioms. The computations validate implementation and expose axiom independence, while the general results remain analytic. The final conclusion is strong but bounded: Resolution Geometry canonically describes distinctions preserved by a declared experiment, and canonically completes their geometry only within a fixed admissibility protocol.

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1 Introduction

A latent state space is normally richer than the distinctions available to one experiment. Treating every latent coordinate as physical, estimable, or geometric overstates what the data identify. Conversely, replacing the latent model by an arbitrary lower-dimensional representation risks discarding distinctions that the experiment does preserve. The correct first object is therefore neither the latent space nor a chosen embedding, but the quotient generated by equality of complete observation laws.

The paper develops the following dependency chain:

$$\boxed{(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow \text{comparison geometry} \longrightarrow P_\tau \longrightarrow W_{\mathrm{obs|adm}} \longrightarrow \text{transport}}$$

and, independently downstream,

$$\boxed{(X, \mu, \pi) \longrightarrow (Q, \pi_*\mu) \longrightarrow H(Q),\ \ H(X \mid Q).}$$

The distinction between the two chains is essential. The quotient determines which distinctions survive. A measure determines how probability is assigned. A protocol determines which resolved directions are accepted as stable, conditioned, transportable, and composable. None of these roles is silently exchanged for another.

Contributions. The strongest theorem ladder established here is:

1. equality-of-law equivalence and universal quotient factorization;
2. maximal operationality and uniqueness up to unique compatible isomorphism;
3. experiment-isomorphism invariance and hidden-extension no-go results;
4. Blackwell-functorial loss of distinctions under garbling;
5. fixed-protocol Fisher, spectral, maximal-sector, and transport characterization;
6. fibrewise representation and exact locus-of-choice theorem;
7. entropy factorization for every operationally invariant state functional;
8. exact Shannon quotient/fibre decomposition and refinement monotonicity;
9. conditional Boltzmann/Gibbs and thermodynamic realization results;
10. deterministic finite attack support and an explicit axiom-independence ledger.

2 Declared experiments and observational equality

Definition 2.1 (General statistical experiment)**.** A declared experiment is

$$\mathcal{E} = (\Theta, (\Omega, \mathcal{A}), \{P_\theta\}_{\theta\in\Theta}),$$

where $(\Omega, \mathcal{A})$ is a measurable outcome space and P_θ is the complete declared observation law. No domination, topology, finite-dimensional parameterization, or smoothness is required at this layer.

Definition 2.2 (Observational equivalence and quotient)**.** Define

$$\theta \sim_{\mathcal{E}} \theta' \quad\iff\quad P_\theta = P_{\theta'}.$$

The observational quotient and projection are

$$Q_{\mathcal{E}} = \Theta/\sim_{\mathcal{E}}, \qquad \pi_{\mathcal{E}} : \Theta \to Q_{\mathcal{E}}, \quad \theta \mapsto [\theta]_{\mathcal{E}}.$$

Definition 2.3 (Experiment-internal construction)**.** A map $F : \Theta \to Z$ is experiment-internal, or observationally extensional, when

$$P_\theta = P_{\theta'} \Longrightarrow F(\theta) = F(\theta').$$

The codomain Z may be a set, measurable space, vector space, tensor space, decision object, or another category whose structure is separately specified.

3 Universal quotient theorem ladder

Theorem 3.1 (Observable quotient factorization)**.** *For every experiment-internal map $F : \Theta \to Z$, there exists a unique map $\widetilde{F} : Q_{\mathcal{E}} \to Z$ satisfying*

$$F = \widetilde{F} \circ \pi_{\mathcal{E}}.$$

Conversely, every composite $\widetilde{F} \circ \pi_{\mathcal{E}}$ is experiment-internal.

Proof. Define $\widetilde{F}([\theta]_{\mathcal{E}}) = F(\theta)$. Extensionality makes this independent of the representative. The displayed factorization follows directly. If two quotient maps give the same factorization, surjectivity of $\pi_{\mathcal{E}}$ makes them equal. The converse follows because $\pi_{\mathcal{E}}$ is constant on equivalence classes. □

Corollary 3.2 (Maximal operational state object)**.** *The quotient retains every distinction made by the complete declared law and removes every distinction that the law cannot make. Equivalently, it is the coarsest quotient through which the complete experiment factors and the finest state object available to all experiment-internal constructions.*

Corollary 3.3 (Universal uniqueness)**.** *Any two objects satisfying the same factorization property are uniquely isomorphic by an isomorphism commuting with their quotient maps.*

Proof. Each candidate factors uniquely through the other. Uniqueness makes both composites identities. □

Theorem 3.4 (Experiment-isomorphism invariance)**.** *An isomorphism of declared experiments preserving all observation laws induces a unique isomorphism of observational quotients. Every natural experiment-internal construction is transported through that isomorphism.*

Proof. Law preservation maps equivalence classes to equivalence classes. Apply quotient factorization to the forward and inverse experiment maps and use uniqueness. □

Theorem 3.5 (Hidden-extension invariance)**.** *Let $\widetilde{\Theta} = \Theta \times Z$ and suppose the declared laws satisfy $\widetilde{P}_{(\theta,z)} = P_\theta$ for all $z \in Z$. Then*

$$\widetilde{\Theta}/\sim_{\widetilde{\mathcal{E}}} \cong \Theta/\sim_{\mathcal{E}} .$$

All natural quotient-level structures agree under this isomorphism.

Proof. Every Z-fibre lies inside one equality-of-law class, while equality between two extended states is exactly equality between their projected laws. □

No-go theorem 3.6 (Latent realization underdetermination)**.** *Equality of complete declared observable laws need not imply isomorphism of latent realizations. Latent multiplicity, hidden dimension, state splitting, parameterization, and invisible topology are not experiment-internal invariants without faithfulness of the observation functor on a declared competitor class.*

Proof. The hidden-extension theorem supplies an immediate family of counterexamples. Finite state-splitting gives another: replace one latent state by several states whose aggregate transitions and emitted observation laws agree with the original. The latent presentations are non-isomorphic although every observable word has the same probability. □

4 Garbling, refinement, and comparison

Definition 4.1 (Blackwell garbling)**.** For experiments on a common parameter set, write $\mathcal{E}_1 \succeq_B \mathcal{E}_2$ when a parameter-independent Markov kernel K satisfies $P_\theta^{(2)} = KP_\theta^{(1)}$ for all θ.

Proposition 4.2 (Canonical quotient morphism under garbling)**.** *If $\mathcal{E}_1 \succeq_B \mathcal{E}_2$, then equality under $\mathcal{E}_1$ implies equality under $\mathcal{E}_2$, and there is a unique surjective map*

$$r_{21} : Q_{\mathcal{E}_1} \to Q_{\mathcal{E}_2}, \qquad \pi_{\mathcal{E}_2} = r_{21} \circ \pi_{\mathcal{E}_1}.$$

Thus garbling can merge observable classes but cannot split an equality-of-law class of the finer experiment.

Proof. If $P^{(1)}_\theta = P^{(1)}_{\theta'}$, applying K yields $P^{(2)}_\theta = P^{(2)}_{\theta'}$. Quotient factorization gives r_{21}; surjectivity follows from surjectivity of both quotient maps. □

Remark 4.3. This is the set-level shadow of Blackwell comparison. Quantitative transfer of metrics or risks requires the corresponding monotonicity and continuity assumptions; quotient universality alone gives no numerical stability modulus.

5 From quotient to fixed-protocol geometry

Definition 5.1 (Admissibility protocol)**.** A protocol is structured data

$$\Gamma = (G, \tau, N, S, C, T),$$

where G is a positive comparison structure; τ is a resolution threshold; N is a nuisance or gauge sector; S denotes stability gates; C denotes conditioning and coarse-graining gates; and T denotes transport, composition, and gluing gates.

Non-claim 5.2 (No protocol selection from quotient logic)*.* The bare experiment does not determine G, τ, N, S, C, or T. The results below are canonical only inside a fixed protocol fibre.

Assumption 5.3 (Regular finite classical fibre)**.** On a regular finite-dimensional classical stratum, assume a smooth distinguishability tensor $F \succeq 0$ satisfying the hypotheses of the classical Fisher–Rao characterization, a declared $G \succ 0$, and a threshold $\tau \notin \operatorname{spec}(G^{-1}F)$.

Theorem 5.4 (Imported Fisher–Rao metric module)**.** *Under sufficient-Markov invariance and the standard regularity hypotheses, the classical local distinguishability metric is Fisher–Rao up to a positive normalization.*

Grounding. This is the Čencov characterization imported only on its stated regular classical domain [5, 6, 7]. It is not a consequence of quotient formation alone. □

Theorem 5.5 (Unique hard resolved projector)**.** *Let $L = G^{-1}F$. The unique G-self-adjoint idempotent commuting with L and implementing the declared upper threshold is*

$$P_\tau = \mathbf{1}_{(\tau,\infty)}(L).$$

Proof. L is self-adjoint in the G inner product. Spectral decomposition reduces any commuting self-adjoint idempotent to binary choices on spectral subspaces. Threshold consistency fixes those choices uniquely. □

Definition 5.6 (Soundness and completeness)**.** Let $R_\tau = \operatorname{im} P_\tau$, and let $\mathsf{Adm}_\Gamma(R_\tau)$ be the declared family of resolved subobjects satisfying every gate. A reported sector W is sound if $W \in \mathsf{Adm}_\Gamma(R_\tau)$ and complete if

$$U \in \mathsf{Adm}_\Gamma(R_\tau) \Longrightarrow U \subseteq W.$$

Lemma 5.7 (Greatest linear gate intersection)**.** *If the gates are linear subspaces $A_N, A_S, A_C, A_T \subseteq R_\tau$, then*

$$W_* = R_\tau \cap A_N \cap A_S \cap A_C \cap A_T$$

is the unique greatest gate-admissible subspace.

Proof. The intersection is contained in every gate and is therefore sound. Every other sound subspace is contained in every gate and hence in the intersection. Two greatest elements contain each other. □

Theorem 5.8 (Maximal admissible-sector characterization). *Whenever the protocol category admits a greatest gate-admissible resolved subobject, soundness and completeness characterize it uniquely:*

$$\boxed{W_{\mathrm{obs|adm}} = \max \mathsf{Adm}_\Gamma(R_\tau).}$$

For the finite linear protocol,

$$\boxed{W_{\mathrm{obs|adm}} = \mathrm{im}\, P_\tau \cap N^\perp \cap S \cap C \cap T.}$$

Corollary 5.9 (No undeclared shrinkage). *A strict subobject $U \subsetneq W_{\mathrm{obs|adm}}$ is sound but incomplete unless the protocol is amended by a gate excluding the omitted directions.*

Theorem 5.10 (Natural transport and gluing). *Suppose protocol-preserving transition maps transport F, G, P_τ and all gates covariantly, satisfy identities, and obey the cocycle law $T_{jk}T_{ij} = T_{ik}$. Then transitions map maximal admissible sectors isomorphically and glue them without path-dependent ambiguity.*

Proof. The image of a sound sector is sound by covariance. The inverse image of any target-admissible sector is source-admissible, so source maximality implies target maximality of the image. The cocycle law identifies direct and composite transport. □

Theorem 5.11 (Fixed-protocol representation). *Under quotient extensionality, the regular classical metric module, separated threshold, hard spectral resolution, existence of a greatest admissible subobject, soundness, completeness, naturality, and cocycle gluing, the object*

$$\mathrm{RG}(\mathcal{E}, \Gamma) = \big(Q_\mathcal{E}, F, G, P_\tau, W_{\mathrm{obs|adm}}, \{T_{ij}\}, \Gamma\big)$$

is determined up to protocol-preserving natural isomorphism and the declared Fisher normalization.

Proof. The quotient universal property fixes $Q_\mathcal{E}$. The imported metric characterization fixes F up to normalization. Spectral functional calculus fixes P_τ. Greatest-subobject maximality fixes $W_{\mathrm{obs|adm}}$. Naturality and the cocycle law fix transported copies and their identifications. Every residual choice is an explicit component of Γ. □

Corollary 5.12 (Exact locus of choice). *Within the theorem's domain, every output component is either forced by a universal, characterization, spectral, greatest-subobject, or gluing property, or explicitly declared in Γ. No third, hidden degree of freedom remains.*

6 Entropy factorization

6.1 State functionals

Definition 6.1 (Operationally invariant functional). A functional $\mathfrak{H} : \Theta \to \mathbb{R}$ is operationally invariant if $\theta \sim_\mathcal{E} \theta'$ implies $\mathfrak{H}(\theta) = \mathfrak{H}(\theta')$.

Theorem 6.2 (Operational entropy factorization). *Every operationally invariant entropy-like state functional factors uniquely:*

$$\mathfrak{H} = \widetilde{\mathfrak{H}} \circ \pi_\mathcal{E}, \qquad \widetilde{\mathfrak{H}} : Q_\mathcal{E} \to \mathbb{R}.$$

Proof. This is the quotient factorization theorem with codomain $\mathbb{R}$. □

Remark 6.3. This theorem is exact but deliberately modest. It does not construct a probability measure, select Shannon entropy among entropy functionals, or identify entropy with thermodynamic entropy. It states only that an experiment-internal state functional cannot vary between representatives the experiment declares identical.

6.2 Random variables and exact quotient/fibre decomposition

Theorem 6.4 (Shannon quotient decomposition)**.** *Let X be a discrete random variable and $Q = \pi(X)$ a deterministic quotient. If the entropies are finite, then*

$$\boxed{H(X) = H(Q) + H(X \mid Q).}$$

Proof. Since Q is a deterministic function of X, $H(X, Q) = H(X)$. The Shannon chain rule gives $H(X, Q) = H(Q) + H(X \mid Q)$ [1, 2]. □

Corollary 6.5 (Observable and fibre information)**.** *$H(Q)$ measures uncertainty among observable classes; $H(X \mid Q)$ measures expected uncertainty among representatives remaining inside the observed class. Neither term alone is generally the total microstate entropy.*

Corollary 6.6 (Uniform fibre formula)**.** *If $X \mid Q = q$ is uniform on a finite fibre of multiplicity $\Omega(q)$, then*

$$H(X \mid Q = q) = \log \Omega(q)$$

and

$$H(X) = H(Q) + \mathbb{E}_Q[\log \Omega(Q)].$$

With $S_G = k_B H(X)$, this becomes

$$S_G = k_B H(Q) + \mathbb{E}_Q[k_B \log \Omega(Q)].$$

Corollary 6.7 (Equality boundary)**.** *$H(Q) = H(X)$ if and only if $H(X \mid Q) = 0$, equivalently X is almost surely determined by Q or π is injective on the support.*

Theorem 6.8 (Refinement redistribution)**.** *If Q_f refines Q_c, so $Q_c = r(Q_f)$, then*

$$H(Q_f) \geq H(Q_c), \qquad H(X \mid Q_f) \leq H(X \mid Q_c),$$

while $H(X)$ remains fixed. More precisely,

$$H(Q_f) = H(Q_c) + H(Q_f \mid Q_c).$$

Proof. Apply the chain rule to the deterministic hierarchy $X \mapsto Q_f \mapsto Q_c$ and use nonnegativity of conditional entropy. □

No-go theorem 6.9 (Quotient Shannon entropy is not Boltzmann fibre entropy)**.** *In general,*

$$H(Q) \neq \log \Omega(q)$$

for an individual macrostate q. The left side is an ensemble average uncertainty over quotient values; the right side is the logarithmic multiplicity of one fibre.

Proof. For a uniform distribution on a finite microstate space of size Ω, let fibre q have size Ω_q. Then $p_q = \Omega_q/\Omega$ and

$$H(Q) = \log \Omega - \sum_q p_q \log \Omega_q,$$

which is generally neither $\log \Omega$ nor any single $\log \Omega_q$. □

7 Conditional thermodynamic realization

Definition 7.1 (Physical macrostate map)**.** Let $(\Omega_{\text{mic}}, \mu)$ be a physical microstate probability space and let $m : \Omega_{\text{mic}} \to M$ be a declared physical macrostate map. Its fibres define the physical coarse-graining.

Theorem 7.2 (Conditional physical coarse-graining correspondence)**.** *If the experiment's observational quotient is measurably isomorphic to the physical macrostate partition,*

$$Q_{\mathcal{E}} \cong M,$$

and the observational distribution is the pushforward of the thermodynamically appropriate measure, then the quotient random variable and physical macrostate random variable have the same probability law and hence the same Shannon entropy.

Proof. A measurable partition isomorphism preserves the pushed-forward probabilities. Shannon entropy depends only on that probability mass function. □

Corollary 7.3 (Boltzmann fibre realization)**.** *Under a microcanonical conditional law uniform on each physical macrostate fibre,*

$$S_B(q) = k_B \log \Omega(q) = k_B H(X \mid Q = q).$$

The ensemble Gibbs entropy decomposes as

$$S_G = k_B H(Q) + \mathbb{E}_Q[S_B(Q)].$$

Non-claim 7.4 (No universal entropy identity)*.* The theorem does not assert that information entropy and thermodynamic entropy are universally identical. The bridge requires both partition alignment and a physically justified measure. Energy, Hamiltonian structure, conserved quantities, reservoir assumptions, and nonequilibrium entropy production are additional physical inputs.

No-go theorem 7.5 (No second law from quotient formation alone)**.** *Quotient formation and Shannon decomposition do not imply monotonic increase of thermodynamic entropy in time.*

Proof. The quotient identity is kinematic and holds at each fixed distribution. A monotonic law requires dynamics plus conditions such as a Markov semigroup, detailed balance, coarse-graining assumptions, or another entropy-production theorem. Reversible microscopic dynamics and arbitrary time-dependent partitions provide immediate counterexamples to unconditional monotonicity. □

8 Numerical attack support

8.1 Exhaustive fixed-protocol battery

The numerical work is separated from the analytic proofs. A six-mode protocol was fixed in a G-orthonormal information eigenbasis:

$$\lambda = (0.18, 0.42, 0.91, 1.37, 2.15, 3.80), \qquad \tau = 1.$$

The canonical threshold mask and final gate intersection were

$$P_\tau = (0, 0, 0, 1, 1, 1), \qquad W_{\text{obs}|\text{adm}} = (0, 0, 0, 1, 0, 0).$$

Every binary projector mask was paired with every binary final-sector mask, giving $2^6 \cdot 2^6 = 4096$ competitors.

Table 1: Exhaustive six-mode survivor ledger.

Constraint applied	Survivors
All binary projector/sector pairs	4096
Fixed-threshold consistency	64
Gate soundness	2
Gate completeness/maximality	1
Transport naturality	1
Independent-product composition	1

Proposition 8.1 (Finite battery result)**.** *Within the fully enumerated six-mode binary hard-spectral competitor class, exactly one pair satisfies threshold consistency, soundness, completeness, natural transport, and product composition, and it equals the analytic construction.*

Proposition 8.2 (Independence witness for completeness)**.** *Soundness alone leaves two sectors: the canonical one-dimensional sector and its zero strict subspace. Completeness removes exactly the undeclared zero-sector shrinkage.*

A supplementary attack sampled 128 random oblique similarities of the canonical projection. None simultaneously satisfied idempotence, G-self-adjointness, commutation with L, and threshold consistency.

Remark 8.3 (Epistemic status of the battery). The enumeration proves only the finite proposition over its declared competitor class. It is not the proof of the general representation theorem. Its functions are implementation validation, full coverage of a finite attack surface, and isolation of missing axioms by explicit counterexample.

8.2 Empirical quotient illustration: UCI HAR

The UCI Human Activity Recognition experiment contains 10,299 windows from 30 volunteers performing six labelled activities; signals were sampled at 50 Hz in 2.56 s windows with 50% overlap, with a subject-disjoint 70/30 volunteer split [13, 14]. This dataset is an appropriate illustration of quotient methodology because activity labels, subject identity, sensor realization, and preprocessing choices define distinct candidate partitions.

No theorem in this paper identifies the activity label as the complete observational quotient. A classifier result only establishes separability relative to a feature map, model class, split, and finite sample. A defensible empirical quotient claim requires, at minimum: (i) a declared complete observation law or tester family; (ii) held-out subject evaluation; (iii) stability under nuisance-preserving perturbations; (iv) explicit search for residual subject information; and (v) uncertainty bounds near rank and threshold changes. The UCI HAR data therefore support a practical attack surface, not a thermodynamic interpretation and not a proof of latent invariance.

9 Established theorem support and dependency ledger

Established result	Force used here	Boundary retained
Quotient universal property	Unique factorization of class-constant maps	Does not select geometry or ontology
Blackwell comparison [3, 4]	Garbling cannot create experiment-relative distinctions	Does not choose a metric, threshold, or protocol

Shannon chain rule [1, 2]	Exact quotient/fibre entropy decomposition	Does not identify thermodynamic entropy
Čencov characterization [5, 6]	Fisher–Rao metric up to scale on its regular classical category	Not valid without its invariance and regularity hypotheses
Spectral theorem and functional calculus [8, 9]	Unique threshold projector after (F, G, τ) is fixed	Does not choose G or τ
Subspace lattice / greatest subobject [10]	Gate intersection is the unique greatest linear admissible sector	Infinite-dimensional closure may require extra hypotheses
Categorical uniqueness [11]	Universal and greatest objects are unique by compatible isomorphism	Not literal equality across different protocol fibres
Boltzmann/Gibbs counting [12]	Physical fibre entropy under ensemble assumptions	Requires a physical coarse-graining and measure

10 Axiom-independence and failure ledger

Removed condition	Surviving ambiguity or failure
Complete declared law	Equality may reflect an incomplete tester family rather than operational identity
Observational extensionality	A claimed observable can vary within one equality-of-law class
Metric characterization hypotheses	Alternative distinguishability geometries survive
Threshold separation	Projector rank can jump and the selected sector is unstable
Threshold consistency	Arbitrary spectral subprojections survive
Gate soundness	Rejected nuisance or unstable directions may enter
Gate completeness	Zero and arbitrary strict sound subspaces survive
Greatest-subobject existence	A unique maximal admissible sector need not exist
Naturality	Coordinate- or presentation-dependent assignments survive
Cocycle closure	Transport depends on overlap path
Probability law	Entropy of quotient values is undefined
Partition alignment	Observable entropy need not be physical macrostate entropy
Thermodynamic measure	Shannon weights need not have thermodynamic meaning
Dynamical assumptions	No entropy-production or second-law conclusion follows
Faithful competitor category	Distinct latent realizations remain observationally identical

11 Non-claims

Non-claim 11.1 (No quotient ontology). The quotient is the maximal operationally distinguishable structure of the declared experiment. It is not asserted to be the complete physical reality.

Non-claim 11.2 (No protocol-free geometry). The experiment fixes observational equivalence, not the admissibility contract. Fibrewise canonicity is not a canonical map $\mathcal{E} \mapsto \Gamma$.

Non-claim 11.3 (No unrestricted Fisher geometry). Fisher–Rao is used only on the regular finite classical module. Singular, non-dominated, infinite-dimensional, and quantum experiments require separate modules.

Non-claim 11.4 (No universal final sector). $W_{\mathrm{obs|adm}}$ is greatest inside one fixed gate family. Different legitimate protocols can produce different final sectors.

Non-claim 11.5 (No latent-dimension identification). Observable quotient dimension or rank does not identify latent dimension because invisible extensions preserve all declared laws.

Non-claim 11.6 (No entropy without a measure). A quotient determines distinguishable classes, not their probabilities. Entropy requires a probability law or measure in addition to the quotient.

Non-claim 11.7 (No $H(Q) = \log \Omega(q)$ shortcut). Quotient Shannon entropy and fibre Boltzmann entropy are distinct except under special degeneracies.

Non-claim 11.8 (No thermodynamics from UCI HAR). The HAR illustration tests operational invariance and nuisance structure only. It carries no thermodynamic identification.

12 Falsification and revision conditions

The framework requires revision in a claimed application if any of the following occurs:

1. an experiment-internal output varies within an equality-of-complete-law class;
2. the purported law is incomplete relative to a tester that distinguishes two quotient representatives;
3. two distinct hard projectors satisfy every fixed selector axiom away from the spectrum;
4. two distinct greatest admissible subobjects exist for the same complete gate family;
5. a strict reported sector is retained without a declared exclusion gate;
6. transport changes the characterized object beyond compatible natural isomorphism;
7. direct and composite overlap transports disagree;
8. a claimed stable sector lacks a positive margin from rank, threshold, or conditioning discriminants;
9. an entropy value is claimed without specifying the law and logarithm convention;
10. thermodynamic identification is claimed without partition alignment and a physical measure;
11. empirical hidden-variable attacks reveal reproducible residual distinctions that the proposed quotient erased.

13 Open problems

Open problem 13.1 (Approximate quotients). Develop a quantitative replacement for equality of laws using Le Cam deficiency, decision metrics, or task-specific tolerances, and prove stability bounds for quotient geometry away from discriminants.

Open problem 13.2 (Infinite and nonlinear protocols). Establish existence and uniqueness of closed greatest admissible subobjects in infinite-dimensional, nonlinear, stratified, and sheaf-valued settings.

Open problem 13.3 (Entropy under stochastic coarse-graining). Extend the deterministic decomposition to declared stochastic channels while separating mutual information, conditional entropy, entropy production, and loss of recoverability.

Open problem 13.4 (Quantum observation systems). Use equality of all declared tester laws for states, channels, instruments, and process tensors; retain the Petz family unless an additional operational principle selects a metric.

Open problem 13.5 (Empirical completeness). Design adversarial finite-sample tests that can reject proposed observational equivalences while honestly acknowledging that equality of unrestricted laws cannot be established from finite data alone.

14 Conclusion

The strongest final quotient construction is the universal equality-of-complete-law quotient

$$Q_{\mathcal{E}} = \Theta/\!\sim_{\mathcal{E}} .$$

Its strength is exact and categorical: every experiment-internal quantity factors through it, any competing object with the same property is uniquely compatible, and invisible latent extensions cannot alter it. Its boundary is equally exact: the quotient does not select an admissibility protocol, probability measure, thermodynamic interpretation, or latent ontology.

Once a regular protocol is fixed, established characterization and spectral results, combined with soundness and completeness, yield the unique maximal observable sector

$$W_{\mathrm{obs|adm}} = \mathrm{im}\, P_{\tau} \cap N^{\perp} \cap S \cap C \cap T$$

within that fibre. The finite attack supports this closure by eliminating all but one of 4096 discrete competitors and by finding no survivor among 128 oblique projector attacks, but the theorem is analytic rather than empirical.

Entropy enters only after a law is placed on the quotient. The exact identity

$$H(X) = H(Q) + H(X \mid Q)$$

then separates observable uncertainty from unresolved fibre multiplicity. Thermodynamic entropy is recovered only when the observational and physical coarse-grainings coincide and the measure is physically appropriate. The final claim is therefore neither ontological nor merely algorithmic:

> Resolution Geometry is the geometry of distinctions that a declared experiment can operationally preserve, together with the canonical maximal admissible structure induced on those distinctions once the complete protocol is fixed. Entropy factorizes through the resulting quotient only after a probability law is supplied; thermodynamic meaning is a further conditional physical realization.

Reproducibility statement

The finite figures reported here correspond to the deterministic fixed-protocol search with seed `20260805`; the exhaustive binary enumeration is seed-independent, while the seed controls the supplementary oblique-projector attack. Numerical artifacts support implementation and independence analysis and are not substitutes for proofs.

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A Machine-readable finite summary

```
all binary projector/sector competitors:  4096
survive threshold consistency:              64
survive gate soundness:                      2
survive gate completeness:                   1
survive natural transport:                   1
survive product composition:                 1
random oblique projectors tested:          128
oblique survivors of all rules:              0
```

B Notation ledger

Symbol	Meaning
$\mathcal{E}$	complete declared experiment
$\theta \sim_{\mathcal{E}} \theta'$	equality of complete declared observation laws

$Q_{\mathcal{E}}$	observational quotient
$\pi_{\mathcal{E}}$	quotient projection
Γ	admissibility protocol
F	local distinguishability/Fisher tensor where defined
G	declared positive comparison structure
P_τ	hard spectral resolved projector
$W_{\mathrm{obs}\mid\mathrm{adm}}$	greatest gate-admissible resolved sector
X	microstate random variable
$Q = \pi(X)$	observable quotient random variable
$H(Q)$	uncertainty among observable classes
$H(X \mid Q)$	expected unresolved fibre uncertainty
$\Omega(q)$	multiplicity of fibre q under a finite uniform conditional law